EPSR-test-ex33 mixato + fmole: intramolecular shaking

Program(s): mixato, fmole | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `fmole`, the EPSR tool [1] that relaxes molecular geometries by intramolecular Monte Carlo [2] before an EPSR run [3,4], following the standard route: build a box with `mixato`, then shake it, and shake a single molecule including rotations of its rotational groups.

Code exercised

Fortran (original, fortran_src/):

- `fmole.f`: `fmole`, `initpt_fmole` (intramolecular potential tables), `update_ato` (refresh from templates)
- `molfit.f`: `molfit` (atom moves, `rotati` internal rotations); `ato_manipulation_rev4.f`: `fmolec1`, `adjust_intra_wts`, `generate_mp`, `makemole`
- `cellneighbourlist.f`: `initialise_cellneighbourlist`; `minimagec.f`: `minimagec`
- `mixato.f`: `mixato`; `ato_read_and_save.f`: `minitc` (read `.ato`), `moloutc` (write `.ato`)

Rust (port, rust/epsr-core/src/):

- `tools/fmole.rs`: `run`, `initpt_fmole`, `update_ato`, `read_template`, `makemole_template`
- `montecarlo.rs`: `molfit`; `energy.rs`: `fmolec1`, `adjust_intra_wts`; `potential.rs`: `generate_mp`
- `neighbour.rs`: `initialise_cellneighbourlist`, `minimagec`; `tools/mixato.rs`: `run`; `ato.rs`: `minitc`, `moloutc`, `ftype`

Physical system

Molecules of the EPSR-test DMSO-water system [5]: a box of 10 DMSO + 20 water (160 atoms) at 0.1 atoms/Å³ and 300 K, cubic side 11.696 Å (`mixbox.ato`), and a single DMSO molecule (`DMSO.ato`, box 21.544 Å) with two methyl rotational groups. `fmole` moves atoms under the intramolecular energy only (harmonic bond and angle restraints plus the repulsive core of `generate_mp`) with a Metropolis test; internal rotations are used only for a single molecule: `fmole.f` sets `stepri = 0` when nmol > 1, since intermolecular forces are not defined in `fmole`.

Inputs

`DMSO.ato` and `WaterMolecule.ato` from EPSR-test and `mixato.stdin` (2 files, 10 DMSO, 20 water, density 0.1, output `mixbox`). Commands: `mixato` < `mixato.stdin`, `fmole mixbox.ato 50` (50 shakes of the 30-molecule box; templates read by `update_ato`), `fmole DMSO 200` (200 shakes with CH3 rotations; the whole-molecule and rotation step sizes are set to zero during `fmole` and restored afterwards, so the step-size line of the `.ato` is unchanged).

How the accuracy was verified

`commands.txt` (3 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Three program calls in one pass. Compared outputs: `mixbox.ato`, the rewritten `DMSO.ato` and `dirlist.txt` — coordinates, step sizes and random-number state are all identical. The shakes are effective: the atom coordinates of `DMSO.ato` in the Fortran reference differ from the input file (e.g. the first atom's offset changes from -0.2807 to -0.2054 Å in x).

Notes

No `.mol` files are present, so the `makemole` call that `fmole` makes for each template does nothing, as in the Fortran; the port stops with an "unsupported" error if a `.mol` file exists.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller and E. Teller, "Equation of state calculations by fast computing machines", J. Chem. Phys. 21, 1087-1092 (1953).
[3] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[4] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.